

Purbanchal University

Faculty of Engineering, Biratnagar, NEPAL

Eighth Semester's Course Structure & Syllabus

Program: Bachelor in Computer Engineering

Effective from 2021 (2078) Batch

Year-IV

Semester-VIII

S.N.	Course code	Subject	Credit Hours	L	T	P	Total	Internal		Final		Total
								Th.	P	Th	P	
1	BCE8028	Cyber Security	3	3	1	2	6	40	25	60	-	125
2	BEG8009	Engineering Professional Practice	2	2	1	-	3	20	-	30	-	50
3	BCE88--	Elective-IV	3	3	1	2	6	40	25	60	-	125
4	BCE8029	Project-III (Part B)	3	1	-	4	5	-	100	-	50	150
5	BCE8030	Internship	3	3	1	-	4		100	-	-	100
		Total	14	12	4	8	24					550

Note-

L: Lecture

T: Tutorial

P: Practical

Th. : Theory



CYBER SECURITY

BCE 8028

Year: IV

Semester: VIII

Teaching Schedule Hours/Week					Examination Schedule						Total Marks
					Final				Internal Assessment		
					Theory		Practical		Theory Marks	Practical Marks	
Credit Hours	L	T	P	Total	Duration	Marks	Duration	Marks	40	25	125
3	3	1	2	6	3 Hrs.	60	-	-			

Note: L: Lecture T: Tutorial P: Practical

Course Objectives:

- To understand cyber threats, vulnerabilities, and attacks
- To learn security mechanisms for networks, systems, and applications
- To apply cryptographic techniques for secure communication
- To understand cyber laws, ethics, and compliance
- To build awareness of emerging cyber security challenges

Course Contents:

Unit 1: Introduction to Cyber Security (7 Hours)

- Introduction to Cyber Space and Cyber Security
- CIA Triad: Confidentiality, Integrity, Availability
- Cyber Security terminology and taxonomy
- Threats, vulnerabilities, risks, and attacks
- Types of attackers: hackers, crackers, insiders, script kiddies
- Cyber kill chain
- Security policies, standards, and best practices



- Overview of security models (Bell-LaPadula, Biba Model)

Unit 2: Cyber Threats and Attack Techniques (8 Hours)

- Malware: viruses, worms, trojans, spyware, ransomware
- Phishing, spear phishing, whaling attacks
- Password attacks: brute force, dictionary, rainbow tables
- Social engineering attacks
- Web-based attacks: SQL injection, XSS, CSRF
- Denial of Service (DoS) and Distributed DoS (DDoS) attacks
- Insider threats
- Case studies of major cyber attacks

Unit 3: Cryptography and Network Security (10 Hours)

3.1 Cryptography:

- Introduction to cryptography and cryptanalysis
- Symmetric key cryptography (DES, AES)
- Asymmetric key cryptography (RSA, Diffie-Hellman, ECC)
- Hash functions (MD5, SHA family)
- Digital signatures and certificates
- Public Key Infrastructure (PKI)

3.2 Network Security

- Secure communication protocols: SSL/TLS, HTTPS
- IP Security (IPSec)
- Firewalls: types and configurations
- Virtual Private Networks (VPNs)



Unit 4: System, Application, and Cloud Security (10 Hours)

4.1 System Security

- Operating system security concepts
- Authentication and authorization mechanisms
- Access control models (DAC, MAC, RBAC)
- Secure boot and trusted computing

4.2 Application Security

- Secure software development lifecycle (SSDLC)
- OWASP Top 10 vulnerabilities
- Code injection and buffer overflow attacks
- Secure coding practices

4.3 Cloud & IoT Security

- Cloud security challenges and threats
- Cloud service models and security (IaaS, PaaS, SaaS)
- IoT security issues and countermeasures

Unit 5: Cyber Laws, Digital Forensics & Emerging Trends (10 Hours)

5.1 Cyber Laws & Ethics

- Cyber ethics and professional responsibilities
- Indian IT Act 2000 and amendments
- Cyber crimes and legal procedures
- Intellectual property rights in cyberspace
- Privacy and data protection laws (GDPR overview)

5.2 Digital Forensics

- Introduction to digital forensics
- Evidence collection and preservation
- Network and system forensics



- Incident response lifecycle

5.3 Emerging Trends

- Artificial Intelligence in Cyber Security
- Blockchain and security
- Zero Trust Architecture
- Future challenges in cyber security

Text Books

1. **William Stallings**, *Cryptography and Network Security*, Pearson
2. **Nina Godbole & Sunit Belapure**, *Cyber Security*, Wiley
3. **Charles P. Pfleeger**, *Security in Computing*, Pearson

Reference Books

- Eric Cole, *Network Security Bible*, Wiley
- Thomas R. Peltier, *Information Security Policies, Procedures, and Standards*, Auerbach
- OWASP Documentation (latest)



Purbanchal University

Model Question 2025

Program: BE Computer **Semester:** VII

Subject: BCE 8028 Cyber Security

Full Marks: 60 **Pass Marks:** 24 **Time:** 3 hrs

Group A: Very Short Questions 4*2=8

1. What is cyber space?
2. What is a brute force attack?
3. What is Digital Signature?
4. What is secure boot?

Group B: Short Questions (Answer any 7 questions) [7*4=28 Marks]

5. Explain different types of attackers in cyber security.
6. Explain the Biba Integrity Model.
7. Describe social engineering attacks with examples.
8. Explain insider threats and their risks to organizations.
9. Explain AES algorithm in brief.
10. Explain SSL/TLS protocol and its importance.
11. explain security concerns in IaaS, PaaS, and SaaS models.
12. Explain cyber ethics and professional responsibilities.

Group C: Long Questions (Answer any three questions) [3*8=24 Marks]

13. Explain DES algorithm and its working.
14. Discuss IoT security issues and countermeasures.
15. Discuss emerging trends in cyber security.
16. Explain the incident response lifecycle in detail. Explain authorization mechanisms. [4+4]



ENGINEERING PROFESSIONAL PRACTICE

BEG 8009

Year: IV

Semester: VIII

Teaching Hours/week				Examination Scheme						Total Marks
				Internal		Final				50
				Theory	Practical	Theory		Practical		
Credit Hours	L	T	P			Duration	Marks	Duration	Marks	
2	2	1	-	20	-	1.5 Hours	30	-	-	

Course Objectives:

To introduce ethical and legal environment in which engineering is practiced.

Course Content:

1.0 Engineering Professionalism:

[6 hrs]

- 1.1 Profession/Professional and Professionalism.
- 1.2 Moral and ethics in engineering profession.
- 1.3 Codes of ethics and guidelines for engineering Profession.
- 1.4 Relationship of the engineering Profession to basic science and technology; Relationship to other Profession
- 1.5 Key roles of engineers in development activities.
- 1.6 Law of ethics in engineering practices.
- 1.7 Fundamental canons for professional engineers.
- 1.8 Individual freedoms and society goals

2.0 History of Engineering Practices:-

[4 hrs]

- 2.1 Concept about society, elements of society and types of society.
- 2.2 History of engineering practice in eastern and western society.

3.0 Engineering Professional Practice in Nepal:

[10 hrs]

- 3.1 Nepal Engineering Council and Nepal Engineering Association.
- 3.2 Engineering Council Act
- 3.3 Contract Law, Elements and types of Contract Law.
- 3.4 Globalization and cross cultural issues
- 3.5 Company Registration



- 3.6 FIDIC
- 3.7 Public Private Partnership (PPP)
- 3.8 Tort liability, Elements of Tort liability and Types of Tort liability.
- 3.9 Business and labor laws.
- 3.10 Relationship to foreign firms working in Nepal.
- 3.11 Trade Union Act

4.0 Professional Practice Sectors in Nepal: [4 hrs]

- 4.1 Job description of engineers working in public sectors
- 4.2 Job description of engineers working in private sectors

5.0 Issues on Engineering Professional Ethics: [6 hrs]

- 5.1 Intellectual property rights: copy rights, Trademark and Patent protection.
- 5.2 Engineers in international development.
- 5.3 Conflict and Dispute management
- 5.4 Corporate social responsibilities

References:

1. Carson Morrison and Philip Hughes, “Professional Engineering Practice- Ethical Aspects”, McGraw- Hill Ryerson Ltd., Toronto 1982
2. Dr. Rajendra Adhikari, “Engineering Professional Practice- Nepalese and International Perspectives” Pashupati Publishing House, Kathmandu Nepal 2010
3. M. Govindarajan; S Natrajan and Senthikumar. “Engineering Ethics”- PHI- Learning Pvt. Ltd. New Delhi 2009

Final Examination Scheme:		
Chapters	Marks	Remarks
1	6	
2	4	
3	10	
4	4	
5	6	
Total	30	
<i>Note: There might be deviation in mark distribution.</i> <i>Mandatory: Evaluation should be based on solving approach and steps.</i>		



PURBANCHAL UNIVERSITY
FIRST SEMESTER FINAL EXAM-2025
(MODEL QUESTION)

LEVEL : B. E. Civil/Computer/Bio-Medical/Elex. Comm & Automation

SUBJECT: Engineering Professional Practice

FULL MARKS : 30

PASS MARKS: 12

TIME : 1.5 hours

Answer All Questions:

Group A:

[2×2=4]

1. Define Patent, Copyright and Trademark.
2. Mention few examples of Corporate Social Responsibilities.

Group B:

[4×4=16]

3. Explain Moral and ethics and describe Law of ethics in detail.
4. Discuss Key roles of Engineers in development activities with suitable examples.
5. Describe Fundamental canons for Professional Engineers.
6. Discuss Profession, Professionalism and features of Professionals.
7. Discuss Engineering Registration process by Nepal Engineering Council and explain Code of ethics for Engineers developed by NEC.

Group C:

[2×5=10]

8. Discuss Tender and Tender Process in detail. Explain Prequalification and Post qualification.

OR

Discuss Tort liability, elements of tort, types of negligence and types of liability with suitable examples.

9. A RCC bridge was designed by the designer on behalf of consultant. This was constructed by the reputed “A” class contractor. After the completion of the Construction, traffic was allowed on the bridge. After six months of operation there were crack in the bridge. A Probe Team was established by Road department. The design procedure was okay, but it was found that the quality of steel metal used was not duly tested. The contractor argued that the procedure of construction was in accordance with the instruction of Engineer and specification. There was also lack of proper supervision by the consultant. The design load for the bridge was 20 Tons. It was also reported that there happen to pass more than 20 Tons vehicles also. The consultant was good friend of contractor. Being a member of Probe Team, what is your judgment on the failure of this bridge?



PROJECT-III (Part B)

BCE8029

Year: IV

Semester: VIII

Teaching Hours/week				Examination Scheme						Total Marks
				Internal		Final				
				Theory	Practical	Theory		Practical		
Credit Hours	L	T	P			Duration	Marks	Duration	Marks	
3	3	-	4	-	100	-	-	-	50	150

Course Objective:

- To provide students with the opportunity to apply theoretical knowledge to real-world computer engineering problems.
- To develop practical skills in designing, implementing, and testing software systems, algorithms, or computational solutions.
- To enhance critical thinking and problem-solving abilities through independent project work.
- To improve students' communication and presentation skills through project reports and formal defenses.
- To foster professional confidence and prepare students for careers in the computer engineering industry or for advanced research.
- To encourage the integration of knowledge across different areas of computer engineering, such as software development, hardware interfacing, and data analysis.

Course Details:

Project III (Part B) is a continuation of Project III (Part A) and focuses on completing, refining, testing, and deploying the proposed system. Students are expected to deliver a fully functional, optimized, and documented solution addressing a real-world problem using modern tools and technologies.

Under the supervision and guidance of member/members of faculty each student is required to carry out an individual or group (Maximum of four students/group) project



which provides opportunities for tackling problem to Computer Engineering and is required to submit a project report.

A project may involve one or more of the following:

- a) Conducting an experimental or practical investigation, such as implementing and testing algorithms, systems, or software prototypes.
- b) Preparing a dissertation that includes a comprehensive literature survey and synthesis of existing research in the relevant domain.
- c) Designing and developing a software system, application, or computational solution for a complex problem.

The project will be carried out under the guidance of one or more faculty members, as deemed beneficial for the student. During the initial phase, faculty may deliver lectures and facilitate discussions to provide direction on the project approach. In the subsequent phase, the student will work independently while consulting the faculty as needed to resolve any challenges.

Students are required to submit a draft report prior to the final submission, allowing the faculty advisor to identify and correct significant errors. The final report must be submitted in duplicate to the Head of the Department. Additionally, students are expected to defend their work through a formal presentation.

Course Workflow

- Refinement of MVP developed in Project A based on feedback.
- Complete system implementation with advanced features.
- Comprehensive testing, optimization, and validation.
- Deployment (local/cloud) and preparation of user documentation.
- Final project report submission (minimum 30 pages).
- Final demonstration and viva-voce examination.

Evaluation Scheme:

The students' work will be evaluated based on their performance in the project throughout the semester, including the quality of their work, presentation and communication skills, and performance during the viva voce or project defense. The Department, in coordination with the project supervisor, will be responsible for evaluating 50% of the total internal marks..



Internal Evaluation (100 Marks)

Parameters	Weightage
MVP Refinement & Enhancement	20%
Implementation & Testing	25%
Documentation & Report Writing	20%
Presentation & Viva-voce	15%
Teamwork & Professionalism	10%
Regularity & Involvement	10%

Final Evaluation (50 Marks)

Parameters	Weightage
Project Completion & Functionality	30%
Innovation & Technical Quality	20%
Final Report & Documentation	20%
Presentation & Viva-voce	20%
Teamwork & Professional Conduct	10%



INTERNSHIP

BCE 8030

Year: IV

Semester: VIII

Teaching Hours/week				Examination Scheme						Total Marks
				Internal		Final				
				Theory	Practical	Theory		Practical		
Credit Hours	L	T	P			Duration	Marks	Duration	Marks	
3	-	-	-	-	100			-	-	100

Note: Internship Duration will be 30 working days.

Course Objectives:

- To provide students with practical, hands-on experience in professional computing and IT environments.
- To expose final-year students to real-world computer engineering projects and industry practices.
- To enable students to apply theoretical and practical knowledge to solve real-world problems.
- To develop professional skills, including teamwork, communication, and adherence to industry ethics.
- To offer opportunities for career exploration and to prepare students for professional roles in the computing and IT industry.
- To cultivate a pragmatic and professional mindset, bridging the gap between academic learning and industry expectations.

Course Details:

The students should primarily acquire knowledge on the following topics:

1. General concepts of the working methods of computer engineering firms, software companies, IT organizations, or hardware-based industries.
2. Client interaction, requirement gathering, discussions, and communication related to project acquisition, system analysis, design, development, testing, deployment, and maintenance.
3. Hands-on experience in various types of computer engineering projects related to software development, hardware design, networking, data analysis, cybersecurity, artificial intelligence, embedded systems, or research-oriented work.
4. Designing, coding, debugging, testing, modification, optimization, and version control of software or hardware systems.



5. Technical documentation, system manuals, project reports, and user documentation preparation.
6. Study and analysis of previously completed or deployed projects, systems, or products.
7. Exposure to real-world working environments such as development labs, data centers, testing facilities, or operational IT setups, and learning how to handle changes in requirements, quality assurance, performance optimization, and security considerations.

Nature of Internship:

The internship is an individual activity. Students should work on various types of computer engineering projects related to software, hardware, networking, system design, or research and development. The internship may be carried out at software companies, IT firms, hardware manufacturing units, startups, research organizations, government or non-government organizations, educational institutions, or other relevant establishments.

The student is responsible for the timely completion of all assigned activities and projects while maintaining professional quality standards. Students must maintain daily records of their work and communicate regularly with the supervisor assigned by the department regarding the progress and status of internship activities. Each student must prepare and submit an individual internship report based on the work performed during the internship period.

Evaluation Criteria

SN	Description	Marks
1	College Supervisor	30
2	Report/Viva/Presentation	40
3	Office Supervisor	30
Total		100

The evaluation and marking will be carried out on the basis of the observation/examination by college supervisor and submitted report/viva/presentation and on the basis of continual observation and examination by office supervisor.



DATA SCIENCE (ELECTIVE IV)

BCE 8822

Year: IV

Semester: VIII

Teaching Schedule Hours/Week					Examination Schedule						Total Marks
					Final				Internal Assessment		
					Theory		Practical		Theory Marks	Practical Marks	
Credit Hours	L	T	P	Total	Duration	Marks	Duration	Marks	40	25	125
3	3	1	2	6	3 Hrs	60	-	-			

Note:

L: Lecture

T: Tutorial

P: Practical

Course Objectives

By the end of this course, students will be able to:

- Understand the data science lifecycle and key concepts.
- Apply statistical and machine learning techniques to analyze data.
- Use tools like Python, Jupyter Notebook, NumPy, Pandas, and Scikit-learn.
- Handle data wrangling, exploration, and visualization.
- Build and evaluate predictive models using real-world datasets.

Course Contents

Unit 1: Introduction to Data Science (3 hours)

- 8.1 What is Data Science?
- 8.2 Roles of a Data Scientist
- 8.3 Applications in real-world (engineering, health, IoT, etc.)
- 8.4 Data Science workflow: CRISP-DM, OSEMN model

Unit 2: Data Collection and Preprocessing (6 hours)

- 2.1 Data types and sources (CSV, APIs, Databases)



- 2.2 Data cleaning: handling missing data, duplicates
- 2.3 Data integration and transformation
- 2.4 Feature engineering and selection

Unit 3: Exploratory Data Analysis (EDA) (6 hours)

- 3.1 Descriptive statistics and summarization
- 3.2 Visualization tools: Matplotlib, Seaborn
- 3.3 Understanding data distribution, correlation, and outliers
- 3.4 Dashboarding with basic tools (e.g., Plotly)

Unit 4: Statistical Foundations for Data Science (6 hours)

- 4.1 Probability distributions
- 4.2 Sampling and hypothesis testing
- 4.3 Correlation vs. causation
- 4.4 Introduction to Bayesian thinking

Unit 5: Machine Learning for Data Science (12 hours)

5.1 Supervised Learning (7 hours)

- 5.1.1 Linear Regression
- 5.1.2 Logistic Regression
- 5.1.3 Decision Trees and Random Forests
- 5.1.4 Model evaluation (Confusion Matrix, Accuracy, Precision, Recall, ROC-AUC)

5.2 Unsupervised Learning (5 hours)

- 5.2.1 Clustering: K-means, Hierarchical
- 5.2.2 Dimensionality Reduction: PCA
- 5.2.3 Applications and evaluation techniques

Unit 6: Working with Real-world Data (6 hours)

- 6.1 Case study: social data application
- 6.2 Project pipeline: from raw data to insights
- 6.3 Deployment basics using Streamlit or Flask
- 6.4 Ethical considerations in data science



Unit 7: Tools and Technologies in Data Science (3 hours)

- 7.1 Jupyter Notebooks
- 7.2 Git and GitHub for version control
- 7.3 Intro to cloud platforms (Google Colab, Kaggle)
- 7.4 Overview of Big Data tools (Spark, Hadoop)

Laboratories: A minimum of **twelve lab exercises** shall be conducted to ensure practical exposure to all major features and components covered in this course.

Recommended Textbooks & Resources

1. “Python for Data Analysis” by Wes McKinney
2. “Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow” by Aurélien Géron
3. “An Introduction to Statistical Learning” by James, Witten, Hastie, Tibshirani (Free PDF)
4. Online Platforms: Kaggle.com, Google Colab



Model Question

PURBANCHAL UNIVERSITY

Program: Bachelor in Computer Engineering

Semester: VIII

Subject: BCE 8822 Data Science (Elective IV)

Full Marks: 60

Pass Marks: 24 **Time:** 3:00 hrs.

Very Short Questions (4x2=8)

1. Define the term Data Science and list any two of its applications.
2. What is the difference between supervised and unsupervised learning?
3. Mention any two data visualization libraries used in Python.
4. What is the purpose of feature engineering in a data science project?

Group- B Short Questions (Answer any 7 Questions)

[7x4=28 Marks]

5. Explain the CRISP-DM model used in the data science process.
6. Differentiate between mean, median, and mode with a simple example.
7. Describe the confusion matrix and explain the significance of its components.
6. What is dimensionality reduction? Explain the role of Principal Component Analysis (PCA).
9. Discuss the steps involved in data cleaning and preprocessing.
10. Explain the use of Jupyter Notebook and Google Colab in data science projects.
11. Describe any two clustering techniques used in unsupervised learning.



Group- C Long Questions (Answer any 3 Questions) [3x8=24 Marks]

12. Consider a dataset of student academic records. Explain how you would perform data collection, preprocessing, EDA, and model building to predict students' success using machine learning.
13. You are given a dataset with missing values, redundant features, and categorical variables. Explain how you would handle this dataset to prepare it for training a classification model.
14. Discuss a complete data science project pipeline using a real-world example (e.g., health prediction, housing prices, or traffic analysis). Include data handling, visualization, modeling, and deployment steps.



NEXT GENERATION NETWORK (ELECTIVE IV)

BCE 8823

Year: IV

Semester: VIII

Teaching Schedule Hours/Week					Examination Schedule						Total Marks
					Final				Internal Assessment		
					Theory		Practical		Theory Marks	Practical Marks	
Credit Hours	L	T	P	Total	Duration	Marks	Duration	Marks	40	25	125
3	3	1	2	6	3 Hrs.	60	-	-			

Note: **L:** Lecture **T:** Tutorial **P:** Practical

Course Objectives

- To understand the evolution of communication networks
- To study architectures and protocols of next generation networks
- To analyze IP-based multimedia and wireless networks
- To explore QoS, security, and mobility management in NGN
- To familiarize students with emerging network technologies

Course Contents:

Unit 1: Introduction to Next Generation Network (NGN)

(9 Hours)

1.1 Evolution of Communication Networks

- Circuit-switched vs packet-switched networks
- PSTN, ISDN, and IP-based networks
- Limitations of traditional networks



1.2 Concept of Next Generation Networks

- Definition and characteristics of NGN
- Convergence of voice, data, and video
- NGN service requirements

1.3 NGN Architecture

- ITU-T NGN functional architecture
- Access, transport, control, and service layers
- Role of softswitch and media gateways

Unit 2: IP-Based Multimedia Networks (9 Hours)

2.1 Multimedia Communication Fundamentals

- Voice over IP (VoIP) principles
- Real-Time Transport Protocol (RTP)
- Session Initiation Protocol (SIP)

2.2 IMS Architecture

- IP Multimedia Subsystem (IMS) overview
- IMS components and interfaces
- Call/session flow in IMS

2.3 Multimedia Services

- Video conferencing and streaming
- QoS requirements for multimedia
- Codec standards and signaling issues

Unit 3: Quality of Service and Traffic Engineering (9 Hours)

3.1 QoS Concepts

- QoS parameters: delay, jitter, packet loss



- Service Level Agreements (SLAs)

3.2 QoS Mechanisms

- Integrated Services (IntServ)
- Differentiated Services (DiffServ)
- Resource Reservation Protocol (RSVP)

3.3 Traffic Engineering

- MPLS architecture and operation
- Traffic shaping and policing
- Congestion control mechanisms

Unit 4: Wireless and Mobile Networks (9 Hours)

4.1 Wireless Access Networks

- Wi-Fi and LTE overview
- Broadband wireless access technologies

4.2 Mobility Management

- Mobile IP concepts
- Handover and roaming mechanisms
- Location management

4.3 Next Generation Wireless

- 4G LTE architecture
- Introduction to 5G networks
- Network slicing and ultra-low latency services

Unit 5: NGN Security, Management & Emerging Trends (9 Hours)

5.1 NGN Security

- Security threats in NGN



- Authentication, authorization, and accounting (AAA)
- IPsec and secure signaling

5.2 Network Management

- Fault, configuration, accounting, performance, security (FCAPS)
- SNMP and network monitoring
- Policy-based network management

5.3 Emerging Trends

- Software Defined Networking (SDN)
- Network Function Virtualization (NFV)
- Cloud networking and edge computing
- Future research challenges in NGN

Recommended Textbooks & References

1. NGN Architectures, Protocols and Services by Toni Janevski
2. IP Communications and Services for NGN by Johnson I. Agbinya
3. MPLS and Next-Generation Networks: Foundations for NGN and Enterprise Virtualization by Azhar Sayeed & Monique Morrow (Cisco Press)
4. R. Perlman, *Interconnections: Bridges, Routers, Switches*, Addison-Wesley
5. William Stallings, *High-Speed Networks and Internetworking*, Pearson
6. James F. Kurose & Keith W. Ross, *Computer Networking: A Top-Down Approach*, Pearson
7. ITU-T NGN Recommendations
8. IEEE and IETF RFCs on NGN, SIP, IMS, MPLS, and 5



PURBANCHAL UNIVERSITY

Model Question 2025

Program: Bachelor in Computer Engineering

Semester: VIII

Subject: BCE 8823 Next Generation Computer Network (Elective IV)

Full Marks: 60

Pass Marks: 24 **Time:** 3:00 hrs.

Group- A Very Short Questions (4x2=8)

1. Define *Next Generation Network (NGN)*.
2. What is the purpose of the *Session Initiation Protocol (SIP)*?
3. List any two Quality of Service (QoS) parameters.
4. What is *Software Defined Networking (SDN)*?

Group- B Short Questions (Answer any 7 Questions)

[7x4=28 Marks]

5. Explain the limitations of traditional circuit-switched networks.
6. Describe the functional architecture of Next Generation Networks.
7. Explain the role of RTP in IP-based multimedia communication.
8. Discuss the architecture and functions of IP Multimedia Subsystem (IMS).
9. Describe security threats in Next Generation Networks.
10. Describe the working of MPLS in traffic engineering.
11. Explain mobility management and handover mechanisms in mobile networks.
12. Discuss the importance of Network Function Virtualization (NFV) in NGN.

Group- C Long Questions (Answer any 3 Questions)

[3x8=24 Marks]

13. Explain the evolution of communication networks leading to Next Generation Networks. Discuss the key characteristics of NGN. [4+4]



14. Explain the architecture of 4G LTE networks. Discuss key features of LTE and describe how LTE supports high data rates and mobility. [4+4]
15. Explain Quality of Service mechanisms in NGN. Discuss Integrated Service with suitable example.
16. Describe the concepts of cloud networking and edge computing in NGN. Explain their benefits, impact on network performance,



COMPUTATIONAL COGNITIVE SCIENCE (ELECTIVE IV)

BCE 8824

Year: IV

Semester: VIII

Teaching Schedule Hours/Week					Examination Schedule						Total Marks
					Final				Internal Assessment		
					Theory		Practical		Theory Marks	Practical Marks	
Credit Hours	L	T	P	Total	Duration	Marks	Duration	Marks	40	25	125
3	3	1	2	6	3 Hrs.	60	-	-			

Note: **L:** Lecture **T:** Tutorial **P:** Practical

Course Objectives

- To understand human cognition from a computational perspective
- To model perception, learning, memory, and decision-making
- To relate cognitive theories with AI and machine learning models
- To apply computational models to real-world cognitive tasks
- To introduce ethical and interdisciplinary aspects of cognitive modeling

Course Contents:

Unit 1: Introduction to Cognitive Science & Computation (9 Hours)

- Cognitive science: scope and interdisciplinary nature
- Human cognition: perception, memory, learning, reasoning
- Computational approach to cognition
- Marr's levels of analysis (computational, algorithmic, implementational)
- Symbolic vs connectionist approaches



- Overview of cognitive architectures (ACT-R, SOAR – introductory)

Unit 2: Perception and Attention Models (9 Hours)

- Visual and auditory perception
- Computational models of perception
- Feature detection and pattern recognition
- Attention mechanisms and selective processing
- Bayesian models of perception
- Relation between perception models and computer vision

Unit 3: Learning and Memory Models (9 Hours)

- Types of memory: sensory, working, long-term memory
- Computational models of memory
- Reinforcement learning and human learning
- Supervised vs unsupervised learning in cognition
- Neural networks as cognitive models
- Forgetting, generalization, and transfer learning

Unit 4: Reasoning, Decision-Making & Language (9 Hours)

- Problem solving and reasoning strategies
- Rule-based and logic-based reasoning models
- Decision-making under uncertainty
- Probabilistic reasoning and Bayesian inference
- Computational models of language processing
- Natural language understanding and cognition

Unit 5: Applications, Ethics & Future Directions (9 Hours)

- Cognitive modeling in AI and robotics



- Human–computer interaction (HCI) and usability modeling
- Cognitive computing and intelligent agents
- Brain-inspired computing and neuromorphic systems
- Ethical issues in cognitive modeling
- Future trends in computational cognitive science

Reference Books

1. Thagard, P., *Mind: Introduction to Cognitive Science*, MIT Press
2. Sun, R., *Cognition and Multi-Agent Interaction*, Cambridge
3. Anderson, J. R., *Cognitive Psychology and Its Implications*
4. Russell & Norvig, *Artificial Intelligence: A Modern Approach*



PURBANCHAL UNIVERSITY

Model Question 2025

Program: Bachelor in Computer Engineering

Semester: VIII

Subject: BCE 8824 Computational Cognitive Science (Elective IV)

Full Marks: 60

Pass Marks: 24 **Time:** 3:00 hrs.

Group- A Very Short Questions (4x2=8)

1. What is meant by computational modeling of cognition?
2. Define selective attention.
3. What is working memory?
4. What is cognitive computing?

Group- B Short Questions (Answer any 7 Questions) (7x4=28)

1. Explain Marr's three levels of analysis with examples.
2. Differentiate between symbolic and connectionist approaches to cognition.
3. Explain Bayesian models of perception.
4. Describe computational approaches to visual perception.
5. Explain reinforcement learning as a model of human learning.
6. Discuss neural networks as models of memory.
7. Explain logic-based reasoning model.
8. Discuss ethical challenges in computational cognitive science.

Group- C Long Questions (Answer any 3 Questions) (3x8=24)

1. Explain decision-making under uncertainty using probabilistic reasoning. (Unit 4)
2. Describe computational models of perception and attention. Explain their relevance to computer vision systems.



3. Explain learning and memory from a computational perspective. Discuss the role of neural networks and reinforcement learning.
4. Discuss applications of computational cognitive science in AI, robotics, and human–computer interaction, along with future trends.



AGILE SOFTWARE DEVELOPMENT (ELECTIVE IV)

BCE 8825

Year: IV

Semester: VIII

Teaching Schedule Hours/Week					Examination Schedule						Total Marks
					Final				Internal Assessment		
					Theory		Practical		Theory Marks	Practical Marks	
Credit Hours	L	T	P	Total	Duration	Marks	Duration	Marks	40	25	125
3	3	1	2	6	3 Hrs.	60	-	-			

Note: **L:** Lecture **T:** Tutorial **P:** Practical

Course Objectives

- To understand the need for Agile in modern software development
- To study Agile values, principles, and frameworks
- To apply Agile practices in planning, development, and testing
- To understand team roles, estimation, and project management
- To evaluate scaling, governance, and real-world Agile adoption

Course Contents:

Unit I: Introduction to Agile Software Development (9 Hours)

- Software crisis and limitations of traditional models
- Agile manifesto and twelve principles
- Agile mindset and values
- Agile vs Waterfall model
- Overview of Agile frameworks (Scrum, XP, Kanban, Lean)
- Benefits and challenges of Agile adoption



Unit II: Scrum Framework (9 Hours)

- Scrum roles: Product Owner, Scrum Master, Development Team
- Scrum artifacts: Product backlog, Sprint backlog, Increment
- Scrum events: Sprint planning, Daily Scrum, Sprint review, Retrospective
- User stories and acceptance criteria
- Estimation techniques: Story points, Planning poker
- Release planning and sprint tracking

Unit III: Extreme Programming (XP) & Agile Practices (9 Hours)

- Values and principles of Extreme Programming
- Pair programming and collective code ownership
- Test-Driven Development (TDD)
- Continuous integration and refactoring
- Simple design and coding standards
- Agile documentation and technical debt

Unit IV: Agile Project Management & Quality (9 Hours)

- Agile project planning and monitoring
- Velocity, burndown and burn-up charts
- Risk management in Agile projects
- Agile testing strategies and automation
- DevOps and Continuous Delivery
- Configuration management in Agile



Unit V: Scaling Agile, Governance & Industry Practices (9 Hours)

- Scaling Agile frameworks (SAFe, LeSS, Scrum@Scale)
- Agile contracts and governance
- Distributed and remote Agile teams
- Agile metrics and performance measurement
- Agile transformation in organizations
- Case studies and future trends in Agile

Recommended Textbooks

1. Schwaber & Sutherland, *The Scrum Guide*
2. Beck, K., *Extreme Programming Explained*
3. Pressman, *Software Engineering: A Practitioner's Approach*
4. Rubin, K. S., *Essential Scrum*



Purbanchal University

Model Question 2025

Program: BE Computer **Semester:** VII

Subject: BCE 8825 Agile Software Development (Elective IV)

Full Marks: 60 **Pass Marks:** 24 **Time:** 3 hrs

Group A: Very Short Questions 4*2=8

1. State any two principles of the Agile Manifesto.
2. What is a Sprint in Scrum?
3. Define Test-Driven Development (TDD).
4. What is meant by Agile scaling?

Group B: Short Questions (Answer any 7 questions) [7*4=28 Marks]

5. Explain the need for Agile methodologies in software development.
6. Explain the challenges of distributed and remote Agile teams.
7. Explain Scrum roles and their responsibilities.
8. Describe the Scrum lifecycle with suitable diagram.
9. Explain velocity and burn-up charts used in Agile project monitoring.
10. Discuss the importance of refactoring in Agile development.
11. Explain burndown charts and velocity in Agile project tracking.
12. Discuss challenges faced during Agile transformation in organizations.



Group C: Long Questions (Answer any three questions) [3*8=24 Marks]

13. Explain the benefits and challenges of adopting Agile methodologies in software development.
14. Describe Scrum events in detail and explain their importance in Agile development.
15. Explain Agile testing strategies and the role of automation in ensuring software quality.
16. Discuss scaling Agile in large organizations with reference to SAFe, and LeSS frameworks.



BLOCK CHAIN (ELECTIVE IV)

BCE 8826

Year: IV

Semester: VIII

Teaching Schedule Hours/Week					Examination Schedule						Total Marks
					Final				Internal Assessment		
					Theory		Practical		Theory Marks	Practical Marks	
Credit Hours	L	T	P	Total	Duration	Marks	Duration	Marks	40	25	125
3	3	1	2	6	3 Hrs.	60	-	-			

Note: L: Lecture T: Tutorial P: Practical

Course Objectives

- To understand the fundamentals and architecture of blockchain systems
- To study cryptographic techniques used in blockchain
- To analyze consensus mechanisms and smart contracts
- To explore public and private blockchain platforms
- To understand real-world blockchain applications and challenges

Unit 1: Introduction to Blockchain (9 Hours)

- Evolution of digital currencies
- Need for blockchain and decentralized systems
- Blockchain definition and characteristics
- Distributed ledger technology
- Blockchain vs traditional databases
- Types of blockchain: public, private, consortium



- Basic blockchain architecture and components

Unit 2: Cryptography & Blockchain Data Structures (9 Hours)

- Cryptographic hash functions
- Digital signatures and public key cryptography
- Merkle trees and block structure
- Hash pointers and immutability
- Wallets, addresses, and key management
- Security properties of blockchain systems

Unit 3: Consensus Mechanisms & Mining (9 Hours)

- Consensus problem in distributed systems
- Proof of Work (PoW)
- Proof of Stake (PoS)
- Delegated Proof of Stake (DPoS)
- Byzantine Fault Tolerance (BFT)
- Mining process and incentives
- Forks and chain reorganization

Unit 4: Smart Contracts & Blockchain Platforms (9 Hours)

- Smart contracts: concepts and lifecycle
- Ethereum architecture
- Solidity basics
- Gas, transactions, and execution model
- Hyperledger Fabric architecture
- Permissioned vs permissionless blockchains



Unit 5: Blockchain Applications, Security & Future Trends (9 Hours)

- Blockchain in finance and cryptocurrencies
- Supply chain management
- Healthcare, identity, and e-governance
- Blockchain security issues and attacks
- Scalability and interoperability challenges
- Legal, ethical, and regulatory issues
- Future trends in blockchain technology

Recommended Textbooks

1. Antonopoulos, *Mastering Bitcoin*, O'Reilly
2. Narayanan et al., *Bitcoin and Cryptocurrency Technologies*, Princeton
3. Drescher, *Blockchain Basics*, Apress
4. Hyperledger & Ethereum official documentation



PURBANCHAL UNIVERSITY

Model Question 2025

Program: Bachelor in Computer Engineering

Semester: VIII

Subject: BCE 8826 Block Chain (Elective IV)

Full Marks: 60

Pass Marks: 24 **Time:** 3:00 hrs.

Group- A Very Short Questions (4x2=8)

1. Define blockchain.
2. What is a cryptographic hash function?
3. What is Proof of Work (PoW)?
4. What is meant by a smart contract?

Group- B Short Questions (Answer any 7 Questions) (7x4=28)

5. Explain the need for blockchain technology.
6. Differentiate between blockchain and traditional databases.
7. Explain the role of Merkle trees in blockchain.
8. Describe the working of digital signatures in blockchain systems.
9. Compare Proof of Work and Proof of Stake consensus mechanisms.
10. Explain the process of mining in blockchain networks.
11. Explain Ethereum architecture and its components.
12. Discuss major security challenges in blockchain systems.

Group- C Long Questions (Answer any 3 Questions) (3x8=24)

13. Discuss blockchain applications in finance, supply chain management, healthcare, and e-governance, along with future trends.



14. Explain cryptographic techniques used in blockchain. Discuss how they ensure security and immutability.
15. Explain different consensus mechanisms used in blockchain networks. Compare their advantages and limitations.
16. Explain the Ethereum architecture.



DIGITAL SOLUTIONS FOR CLIMATE CHANGE (ELECTIVE IV)

BCE 8827

Year: IV

Semester: VIII

Teaching Schedule Hours/Week					Examination Schedule						Total Marks
					Final				Internal Assessment		
					Theory		Practical		Theory Marks	Practical Marks	
Credit Hours	L	T	P	Total	Duration	Marks	Duration	Marks	40	25	125
3	3	1	2	6	3 Hrs.	60	-	-			

Note: L: Lecture T: Tutorial P: Practical

Course Objectives:

- To understand climate change challenges from a technological perspective
- To explore digital technologies for climate mitigation and adaptation
- To apply computing solutions to environmental sustainability problems
- To encourage interdisciplinary and socially responsible engineering design

Course Outcomes:

After completing this course, students will be able to:

- Explain climate change issues and the role of digital technologies
- Apply data analytics and AI for climate monitoring and prediction
- Design IoT-based environmental monitoring systems
- Evaluate digital solutions for sustainable energy and smart cities
- Propose innovative computing solutions for climate resilience

Course Contents:

Unit 1: Climate Change & Role of Digital Technologies (9 Hours)



1.1 Introduction to Climate Change

- Climate change science: causes, impacts, global trends
- Greenhouse gases and carbon footprint
- Climate mitigation vs adaptation strategies

1.2 Digital Transformation for Sustainability

- Role of computer engineering in climate action
- Digital sustainability and green computing
- Climate informatics and environmental data science

1.3 Global Initiatives & Case Studies

- UN SDGs and climate goals
- Digital climate solutions: global case studies
- Role of governments, startups, and tech companies

Unit 2: Data Analytics & AI for Climate Solutions (9 Hours)

2.1 Climate Data Collection & Processing

- Climate data sources: satellites, sensors, open datasets
- Data preprocessing and visualization techniques

2.2 Machine Learning Applications

- Climate prediction and forecasting models
- AI for weather, flood, and drought prediction
- Image processing for satellite and remote sensing data

2.3 Decision Support Systems

- AI-driven climate risk assessment
- Early warning systems
- Ethical AI and sustainability considerations



Unit 3: IoT & Smart Environmental Monitoring (9 Hours)

3.1 IoT Fundamentals for Climate Monitoring

- Sensors for temperature, air quality, water, and soil
- IoT architecture and communication protocols

3.2 Smart Environmental Systems

- Smart agriculture and precision farming
- Air and water pollution monitoring systems
- Forest fire detection and disaster management

3.3 Edge Computing & Energy Efficiency

- Edge vs cloud computing for environmental data
- Low-power IoT devices and green IoT
- Challenges in deployment and scalability

Unit 4: Digital Solutions for Energy & Smart Cities (9 Hours)

4.1 Smart Energy Systems

- Renewable energy monitoring systems
- Smart grids and energy management platforms
- AI-based demand forecasting

4.2 Smart Cities & Sustainable Infrastructure

- Intelligent transportation systems
- Traffic optimization and emission reduction
- Waste management and water conservation systems

4.3 Blockchain & Emerging Technologies

- Blockchain for carbon credit tracking
- Digital twins for sustainable cities



- AR/VR for climate education and awareness

Unit 5: Green Software, Policy & Future Trends (9 Hours)

5.1 Green Computing & Software Engineering

- Energy-efficient algorithms
- Sustainable software design principles
- Carbon-aware computing

5.2 Policy, Security & Ethics

- Climate data security and privacy
- Technology policy and regulatory challenges
- Ethical issues in climate technologies

5.3 Future Trends & Innovation

- AI for climate resilience
- Climate tech startups and innovation ecosystems
- Student project ideation and proposal development

Recommended Textbooks & References

1. Bill Tomlinson, *Greening Through IT*, MIT Press
2. V. Devadasan, *Green Computing*, Wiley
3. IPCC Assessment Reports (latest editions)
4. **Climate University's Portal:** <https://www.climateuniversity.fi/climate-now/>
5. NASA & NOAA Climate Data Portals
6. Research articles from IEEE, ACM on Climate Informatics



PURBANCHAL UNIVERSITY

Model Question 2025

Program: Bachelor in Computer Engineering

Semester: VIII

Subject: BCE 8827 Digital Solutions for Climate Change (Elective IV)

Full Marks: 60

Pass Marks: 24 **Time:** 3:00 hrs.

Group- A Very Short Questions (4x2=8 Marks)

1. Define climate mitigation and climate adaptation.
2. What is meant by climate informatics?
3. List any two environmental parameters commonly monitored using IoT sensors.
4. What is green computing?

Group- B Short Questions (Answer any 7 Questions)

[7x4=28 Marks]

5. Explain the role of digital technologies in addressing climate change challenges.
6. Discuss the contribution of computer engineers towards achieving UN Sustainable Development Goals related to climate action.
7. Describe different sources of climate data used for analytics and prediction.
8. Explain how machine learning can be applied for weather or climate forecasting.
9. Explain the architecture of an IoT-based environmental monitoring system with a neat diagram.
10. Discuss the applications of IoT in smart agriculture and precision farming.
11. Explain the concept of smart grids and their importance in sustainable energy management.
12. Discuss the importance of energy-efficient algorithms in sustainable software development.



Group- C Long Questions (Answer any 3 Questions)

[3x8=24 Marks]

13. Explain in detail the use of AI and data analytics for climate risk assessment and early warning systems.
14. Describe the design and working of an IoT-based air or water pollution monitoring system. Discuss challenges in real-world deployment.
15. Explain how digital solutions contribute to the development of smart and sustainable cities. Illustrate with suitable examples.
16. Discuss green software engineering principles, ethical issues, and future trends in digital climate solutions.



AUTOMATION AND ROBOTICS (ELECTIVE IV)

BCE 8828

Year: IV

Semester: VIII

Teaching Schedule Hours/Week					Examination Schedule						Total Marks
					Final				Internal Assessment		
					Theory		Practical		Theory Marks	Practical Marks	
Credit Hours	L	T	P	Total	Duration	Marks	Duration	Marks	40	25	125
3	3	1	2	6	3 Hrs.	60	-	-			

Note: L: Lecture T: Tutorial P: Practical

Course Objectives

- To understand fundamentals of automation and robotic systems
- To study components, sensors, actuators, and controllers used in robots
- To learn kinematics and control of robotic manipulators
- To explore industrial automation and intelligent robotics
- To understand real-world applications and future trends in robotics

Unit 1: Introduction to Automation and Robotics (9 Hours)

- Definition and scope of automation
- Types of automation: fixed, programmable, flexible
- Evolution of robotics
- Classification of robots
- Components of a robotic system
- Degrees of freedom and robot configurations
- Applications of automation and robotics



Unit 2: Sensors, Actuators, and Controllers (9 Hours)

- Role of sensors in robotics
- Position, proximity, force, vision, and tactile sensors
- Actuators: electric, hydraulic, pneumatic
- Stepper motors and servo motors
- Controllers and control architectures
- PLC fundamentals and applications

Unit 3: Robot Kinematics and Control (9 Hours)

- Coordinate systems and transformations
- Forward kinematics
- Inverse kinematics
- Workspace and singularities
- Motion planning and trajectory generation
- Open-loop and closed-loop control systems

Unit 4: Industrial Automation and Intelligent Robotics (9 Hours)

- Industrial robot programming methods
- Computer-integrated manufacturing (CIM)
- SCADA and distributed control systems
- Mobile robots and autonomous navigation
- Introduction to AI in robotics
- Human–robot interaction

Unit 5: Applications, Ethics & Future Trends (9 Hours)

- Robotics in manufacturing and logistics
- Service robots and medical robotics



- Agricultural and defense robots
- Robotics in smart cities
- Safety, ethics, and societal impact
- Industry 4.0 and future trends in automation

Recommended Textbooks

1. Groover, M. P., *Automation, Production Systems, and Computer-Integrated Manufacturing*
2. Craig, J. J., *Introduction to Robotics: Mechanics and Control*
3. Saeed B. Niku, *Introduction to Robotics*
4. IEEE Robotics & Automation Magazine



PURBANCHAL UNIVERSITY

Model Question 2025

Program: Bachelor in Computer Engineering

Semester: VIII

Subject: BCE 8828 Automation and Robotics (Elective IV)

Full Marks: 60

Pass Marks: 24 **Time:** 3:00 hrs.

Group- A Very Short Questions (4x2=8)

1. Define automation.
2. What is the function of a servo motor?
3. What is meant by forward kinematics?
4. What is Industry 4.0?

Group- B Short Questions (Answer any 7 Questions)

[7x4=28 Marks]

5. Explain different types of automation.
6. Describe various robot configurations with examples.
7. Explain the role of sensors in robotic systems.
8. Compare stepper motors and servo motors.
9. Explain inverse kinematics of a robotic manipulator.
10. Describe motion planning and trajectory generation.
11. Explain the role of PLCs in industrial automation.
12. Discuss ethical and safety issues in robotics.



Group- C Long Questions (Answer any 3 Questions)

[3x8=24 Marks]

13. Explain computer-integrated manufacturing (CIM).
14. Explain different types of sensors and actuators used in robotics with suitable examples.
15. Explain robot kinematics in detail. Discuss forward and inverse kinematics with neat diagrams. [4+4]
16. Discuss applications of robotics in manufacturing, healthcare, and smart cities, along with future trends in automation.

